two modes scheme with frequencies $\omega_0 \pm \omega_m$ and usage of the pump with frequency ω_0 located in between of the modes — see Fig. 11. The system has two degenerate modes. If the position of a perfectly reflecting mirror M is fixed, one MSI mode, characterized with frequency ω_+, is given by a light wave which travels between M_1 and BS. The light is split on the BS and after reflection from mirror M returns exactly to M_1. It does not propagate to M_2. For the other mode, characterized with frequency ω_-, wave travels from M_2 to BS and after reflection from M returns to M_2 and does not propagate to M_1. The frequencies of the modes, $\omega_\pm$, are controlled by variation of path distances $\ell_1,\ \ell_2$. Small shift of mirror M position provides coupling between the modes. Mirror M is a test mass of the mechanical oscillator with mass m and eigenfrequency ω_m. The back action can be suppressed in this scheme as well. However, pump on frequency ω_0 is not resonant and more optical power will be needed to beat the SQL. In addition pump should be excited though both mirrors $M_1,\ M_2$.

We hope that analyzed here two photon nondegenerate squeezing in broadband coherent multidimensional variational measurement can be used in precision optomechanical measurements including laser gravitational wave detectors.

ACKNOWLEDGMENTS

The research of SPV has been supported by by Theoretical Physics and Mathematics Advancement Foundation "BASIS" (Contract No. 22-1-1-47-1), by the Interdisciplinary Scientific and Educational School of Moscow University "Fundamental and Applied Space Research" and by the TAPIR GIFT MSU Support of the California Institute of Technology. This document has LIGO number P2300325.

Appendix A: Description of relaxation

In this appendix we provide details of the standard calculation for intracavity fields, for example, see [47].

We begin with Hamiltonian (2.2)

$$H_{T,\,0} = \sum_{k=0}^{\infty} \hbar\omega_k a_k^\dagger a_k, \tag{A1}$$

$$H_{\gamma_0} = i\hbar\sqrt{\frac{\gamma_0\Delta\omega}{\pi}} \sum_{k=0}^{\infty} \left[(c_+^\dagger + c_-^\dagger)a_k - (c_+ + c_-)a_k^\dagger\right],$$

$$H_{T,\,e} = \sum_{k=0}^{\infty} \hbar\omega_k e_k^\dagger e_k, \tag{A2}$$

$$H_{\gamma_e} = i\hbar\sqrt{\frac{\gamma_e\Delta\omega}{\pi}} \sum_{k=0}^{\infty} \left[(c_+^\dagger + c_-^\dagger)e_k - (c_+ + c_-)e_k^\dagger\right],$$

$$H_{T,\,m} = \sum_{k=0}^{\infty} \hbar\omega_k q_k^\dagger q_k, \tag{A3}$$

$$H_{\gamma_m} = i\hbar\sqrt{\frac{\gamma_m\Delta\omega}{\pi}} \sum_{k=0}^{\infty} \left[d^\dagger q_k - d q_k^\dagger\right]. \tag{A4}$$

Here $H_{T,\,0}$ is the Hamiltonian of the environment presented as a bath of oscillators [2] described with frequencies $\omega_k = \omega_{k-1} + \Delta\omega$ and annihilation and creation operators $a_k,\ a_k^\dagger$. H_{γ_0} is the Hamiltonian of coupling between the environment and the optical resonator, γ_0 is the coupling constant. Physical sense of a_k are amplitudes of modes falling on input mirror of cavity. $H_{T,\,e}$ and H_{γ_e} are the Hamiltonians described another environment and interaction with resonator corresponding to optical losses. Similarly $H_{T,\,m}$ is the Hamiltonian of the environment presented by a thermal bath of mechanical oscillators with frequencies $\omega_k = \omega_{k-1} + \Delta\omega$ and amplitudes described with annihilation and creation operators $q_k,\ q_k^\dagger$. H_{γ_m} is the Hamiltonian of coupling between the environment and the mechanical oscillator, $2\gamma_m$ is the decay rate of the oscillator.

As a first step we account only $H_{T,\,0},\ H_{\gamma_0}$. Heisenberg equations for operators c_+ and a_k are the following:

$$\dot{c}_+ = -i\omega_+ c_+ - \eta c_0 d + \sqrt{\frac{\gamma_0\Delta\omega}{\pi}} \sum_{k=0}^{\infty} a_k, \tag{A5a}$$

$$\dot{a}_k = -i\omega_k a_k - \sqrt{\frac{\gamma_0\Delta\omega}{\pi}}\,(c_+ + c_-) \tag{A5b}$$

We introduce slow amplitudes $c_\pm \to c_\pm e^{-i\omega_\pm t}$, $d \to d e^{-i\omega_m t}$, $a_k \to a_k e^{-i\omega_k t}$ and substitute them into (A5)

$$\dot{c}_+ = -\eta c_0 d + \sqrt{\frac{\gamma_0\Delta\omega}{\pi}} \sum_{k=0}^{\infty} a_k e^{-i(\omega_k - \omega_+)t}, \tag{A6a}$$

$$\dot{a}_k = -\sqrt{\frac{\gamma_0\Delta\omega}{\pi}}\left(c_+ e^{-i\omega_+ t} + c_- e^{-i\omega_- t}\right) e^{i\omega_k t} \tag{A6b}$$

Using initial condition $a_k(t=0) = a_k(0)$ to integrate (A6b) we derive

$$a_k(t) = a_k(0) - \int_0^t \sqrt{\frac{\gamma_0\Delta\omega}{\pi}} c_+(s) e^{-i(\omega_+ - \omega_k)s} ds - \tag{A7}$$

$$- \int_0^t \sqrt{\frac{\gamma_0\Delta\omega}{\pi}} c_-(s) e^{-i(\omega_- - \omega_k)s} ds \tag{A8}$$

Using the condition $a_k(t=\infty) = a_k(\infty)$ to integrate (A6b) we derive

$$a_k(t) = a_k(\infty) + \int_t^\infty \sqrt{\frac{\gamma_0\Delta\omega}{\pi}} c_+(s) e^{-i(\omega_+ - \omega_k)s} ds + \tag{A9}$$

[2] Here and below we consider the equidistant spectrum for all thermal baths.

$$+\int_t^\infty \sqrt{\frac{\gamma_0\Delta\omega}{\pi}}c_-(s)e^{-i(\omega_- -\omega_k)s}ds \tag{A10}$$

To get the input-output relation we substitute initial condition (A8) into (A6a)

$$\dot{c}_+ = -\eta c_0 d + \sum_{k=0}^{\infty}\sqrt{\frac{\gamma_0\Delta\omega}{\pi}}a_k(0)e^{-i(\omega_k-\omega_+)t} - \tag{A11a}$$

$$-\sum_{k=0}^{\infty}\int_0^t \frac{\gamma_0\Delta\omega}{\pi}c_+(s)e^{-i(\omega_k-\omega_+)(t-s)}ds - \tag{A11b}$$

$$-\left(\sum_{k=0}^{\infty}\int_0^t \frac{\gamma_0\Delta\omega}{\pi}c_-(s)e^{-i(\omega_k-\omega_-)(t-s)}ds\right)e^{i(\omega_+-\omega_-)t}$$

and omit the last term proportional to $e^{i(\omega_+-\omega_-)t}$ as the fast oscillating, and define the input field

$$a_+(t) = \sum_{k=0}^{\infty}\sqrt{\frac{\Delta\omega}{2\pi}}a_k(0)e^{-i(\omega_k-\omega_+)t} \tag{A12}$$

To calculate the remaining sum in (A11) we assume the validity of limit $\Delta\omega \to 0$ and replace the sum by the integration

$$\sum_{k=0}^{\infty}\int_0^t \frac{\gamma_0\Delta\omega}{\pi}c_+(s)e^{-i(\omega_k-\omega_+)(t-s)}ds \to \tag{A13a}$$

$$\to \int_0^\infty\int_0^t 2\gamma_0 c_+(s)e^{-i(\omega-\omega_+)(t-s)}ds\frac{d\omega}{2\pi} =$$

$$= \int_{-\omega_+}^\infty\int_0^t 2\gamma_0 c_+(s)e^{-i\omega(t-s)}ds\frac{d\omega}{2\pi} \approx$$

$$\approx \int_{-\infty}^\infty\int_0^t 2\gamma_0 c_+(s)e^{-i\omega(t-s)}ds\frac{d\omega}{2\pi} =$$

$$= \int_0^t 2\gamma_0 c_+(s)\delta(t-s)ds = \frac{2\gamma_0 c_+(t)}{2} = \gamma_0 c_+(t) \tag{A13b}$$

Substituting (A12) and (A13) into (A11) we obtain

$$\dot{c}_+ = -\eta c_0 d + \sqrt{2\gamma_0}a_+ - \gamma_0 c_+. \tag{A14}$$

Taking into account Hamiltonians $H_{T,e}$, H_{γ_e} by similar way we obtain generalization of (A14)

$$\dot{c}_+ = -\eta c_0 d + \sqrt{2\gamma_0}a_+ + \sqrt{2\gamma_e}e_+ - (\gamma_0+\gamma_e)c_+. \tag{A15}$$

By analogue, we derive the equation for input field a_- and present it in a similar form

$$a_-(t) = \sum_{k=0}^{\infty}\sqrt{\frac{\Delta\omega}{2\pi}}a_k(0)e^{-i(\omega_k-\omega_-)t} \tag{A16}$$

It leads to the equation for the intracavity field c_-

$$\dot{c}_- + \gamma_0 c_- - \eta c_0 d^\dagger = \sqrt{2\gamma_0}a_-. \tag{A17}$$

with generalization with account $H_{T,e}$, H_{γ_e}: (A14)

$$\dot{c}_- = -\eta c_0 d^\dagger + \sqrt{2\gamma_0}a_- + \sqrt{2\gamma_e}e_- - (\gamma_0+\gamma_e)c_-. \tag{A18}$$

Similar equation can be derived for the amplitude $q(t)$ of the mechanical oscillator

$$q(t) = \sum_{k=0}^{\infty}\sqrt{\frac{\Delta\omega}{2\pi}}b_{m,k}(0)e^{-i(\omega_k-\omega_+)t}, \tag{A19}$$

resulting in the Langevin equation for mechanical oscillator quadrature d

$$\dot{d} + \gamma_m d - \eta^*[c_0 c_-^\dagger + c_+ c_0^\dagger] = \sqrt{2\gamma_m}q. \tag{A20}$$

To derive the input-output relation we substitute (A10) into (A6a) and define the output amplitudes

$$b_+(t) = -\sum_{k=0}^{\infty}\sqrt{\frac{\Delta\omega}{2\pi}}a_k(\infty)e^{-i(\omega_k-\omega_+)t} \tag{A21}$$

$$b_-(t) = -\sum_{k=0}^{\infty}\sqrt{\frac{\Delta\omega}{2\pi}}a_k(\infty)e^{-i(\omega_k-\omega_+)t}. \tag{A22}$$

It leads to

$$\dot{c}_+ - (\gamma_0+\gamma_e)c_+ + \eta c_0 d = \sqrt{2\gamma_0}b_+ \tag{A23}$$

$$\dot{c}_- - (\gamma_0+\gamma_e)c_- + \eta^* c_0 d^\dagger = \sqrt{2\gamma_0}b_- \tag{A24}$$

Utilizing pairs of equations (A14) and (A23) as well as (A17) and (A24) we obtain the final expression for the input-output relations

$$b_+ = -a_+ + \sqrt{2\gamma_0}c_+ \tag{A25}$$

$$b_- = -a_- + \sqrt{2\gamma_0}c_- \tag{A26}$$

Let us to derive the commutation relations for the Fourier amplitudes of the operators. We introduce Fourier transform of field $a_+(t)$ using (A12)

$$a_+(\Omega) = \int_{-\infty}^{\infty}\sum_{k=0}^{\infty}\sqrt{\frac{\Delta\omega}{2\pi}}a_k(0)e^{-i(\omega_k-\omega_+-\Omega)t}dt = \tag{A27a}$$

$$= \sum_{k=0}^{\infty}\sqrt{2\pi\Delta\omega}a_k(0)\delta(\Omega-\omega_k+\omega_+) \tag{A27b}$$

This allows us to find the commutators (3.8):

$$[a_+(\Omega), a_+^\dagger(\Omega')] = \sum_{k=0}^{\infty}2\pi\Delta\omega[a_k(0), a_k^\dagger(0)]\times$$

$$\times\,\delta(\Omega-\omega_k+\omega_+)\delta(\Omega'-\omega_k+\omega_+) \to$$

$$\to \int_{-\infty}^{\infty} 2\pi[a_+(0), a_+^\dagger(0)]\delta(\Omega-\omega)\delta(\Omega'-\omega)d\omega =$$
$$= 2\pi\delta(\Omega-\Omega'), \tag{A28a}$$

and the correlators (B2) (we assume that oscillators of environment are in thermal state, for optics — in vacuum state)

$$\langle a_+(\Omega), a_+^\dagger(\Omega')\rangle = \sum_{k=0}^{\infty} 2\pi\Delta\omega\langle a_k(0), a_k^\dagger(0)\rangle\times$$
$$\times\,\delta(\Omega-\omega_k+\omega_+)\delta(\Omega'-\omega_k+\omega_+)\to$$
$$\to \int_{-\infty}^{\infty} 2\pi\langle a_+(0)a_+^\dagger(0)\rangle\delta(\Omega-\omega)\delta(\Omega'-\omega)d\omega =$$
$$= 2\pi\delta(\Omega-\Omega') \tag{A29a}$$

Similar expressions can be derived for commutators and correlators of the optical a_- and mechanical q quantum amplitudes.

Appendix B: Input-output relations (two photon squeezing)

Here we present detail formulas for derivation of output quadratures (3.15) for case of two photon squeezing.

Using definition (3.7) and (3.2) we derive commutators and correlators for the Fourier amplitudes of the loss fluctuation operators $e_\pm$ and thermal noise operators $\hat{q}$

$$\left[e_\pm(\Omega), e_\pm^\dagger(\Omega')\right] = 2\pi\,\delta(\Omega-\Omega'), \tag{B1}$$
$$\left\langle e_\pm(\Omega)\, e_\pm^\dagger(\Omega')\right\rangle = 2\pi\,\delta(\Omega-\Omega'), \tag{B2}$$
$$\left[q(\Omega),\, q^\dagger(\Omega')\right] = 2\pi\,\delta(\Omega-\Omega'), \tag{B3}$$
$$\left\langle q(\Omega)\, q^\dagger(\Omega')\right\rangle = 2\pi\left(2n_T+1\right)\delta(\Omega-\Omega'). \tag{B4}$$

For operator $(a_\pm(\Omega))$ of input fluctuations formulas are similar.

Using (3.6) one can derive set for the Fourier amplitudes $c_\pm$ for the intracavity fields as well as mechanical amplitude d:

$$(\gamma-i\Omega)c_+(\Omega) = -\eta C_0 d(\Omega) + \kappa c_-^\dagger(-\Omega)+ \tag{B5a}$$
$$+\sqrt{2\gamma_0}\,a_+(\Omega) + \sqrt{2\gamma_e}e_+(\Omega),$$
$$(\gamma-i\Omega)c_-(\Omega) = \eta C_0 d^\dagger(-\Omega) + \kappa c_+^\dagger(-\Omega)+ \tag{B5b}$$
$$+\sqrt{2\gamma_0}\,a_-(\Omega) + \sqrt{2\gamma_e}\,e_-(\Omega),$$
$$(\gamma_m-i\Omega)d(\Omega) = \eta C_0\left[c_-^\dagger(-\Omega)+c_+(\Omega)\right]+ \tag{B5c}$$
$$+\sqrt{2\gamma_m}\,\hat{q}(\Omega) + f_s(\Omega).$$

Let introduce quadratures of amplitude and phase (3.13). Then using (B5) we obtain

$$(\gamma-i\Omega)c_{+a} - \kappa c_{-a} + \eta C_0 d_a = \tag{B6a}$$
$$= \sqrt{2\gamma_0}a_{+a} + \sqrt{2\gamma_e}e_{+a},$$
$$(\gamma-i\Omega)c_{+\phi} + \kappa c_{-\phi} + \eta C_0 d_\phi = \tag{B6b}$$
$$+\sqrt{2\gamma_0}a_{+\phi} + \sqrt{2\gamma_e}e_{+\phi},$$
$$(\gamma-i\Omega)c_{-a} - \kappa c_{+a} - \eta C_0 d_a = \tag{B6c}$$
$$= \sqrt{2\gamma_0}a_{-a} + \sqrt{2\gamma_e}e_{-a},$$
$$(\gamma-i\Omega)c_{-\phi} + \kappa c_{+\phi} + \eta C_0 d_\phi = \tag{B6d}$$
$$= \sqrt{2\gamma_0}a_{-\phi} + \sqrt{2\gamma_e}e_{-\phi},$$
$$(\gamma_m-i\Omega)d_a - \eta C_0\Big(c_{+a}+c_{-a}\Big) = \tag{B6e}$$
$$= \sqrt{2\gamma_m}q_a + f_{s\,a},$$
$$(\gamma_m-i\Omega)d_\phi - \eta C_0\Big(c_{+\phi}-c_{-\phi}\Big) = \tag{B6f}$$
$$= \sqrt{2\gamma_m}q_\phi + f_{s\,\phi}.$$

Please note that sum $(c_{+a}+c_{-a})$ does not contain information on the mechanical motion (term proportional to $\sim d_a$ is absent), but produces the back action term in (B6e). At the same time difference $(c_{+\phi}-c_{-\phi})$ does not contain information on mechanical motion (term $\sim d_\phi$), but is responsible on back action in (B6f). So it should be useful to introduce sum and difference of the quadratures (3.14a) and rewrite (B6) in the new notations

$$(\gamma-\kappa-i\Omega)g_{a+} = \sqrt{2\gamma_0}\alpha_{a+} + \sqrt{2\gamma_e}\epsilon_{a+}, \tag{B7a}$$
$$(\gamma+\kappa-i\Omega)g_{a-} + \sqrt{2}\eta C_0 d_a = \sqrt{2\gamma_0}\alpha_{a-}+ \tag{B7b}$$
$$+\sqrt{2\gamma_e}\epsilon_{a-},$$
$$(\gamma_m-i\Omega)d_a - \sqrt{2}\eta C_0 g_{a+} = \sqrt{2\gamma_m}q_a + f_{s\,a}, \tag{B7c}$$
$$(\gamma+\kappa-i\Omega)g_{\phi+} + \sqrt{2}\eta C_0 d_\phi = \sqrt{2\gamma_0}\alpha_{\phi+}+ \tag{B7d}$$
$$+\sqrt{2\gamma_e}\epsilon_{\phi+},$$
$$(\gamma-\kappa-i\Omega)g_{\phi-} = \sqrt{2\gamma_0}\alpha_{\phi-} + \sqrt{2\gamma_e}\epsilon_{\phi-}, \tag{B7e}$$
$$(\gamma_m-i\Omega)d_\phi - \sqrt{2}\eta C_0 g_{\phi-} = \sqrt{2\gamma_m}q_\phi + f_{s\,\phi}. \tag{B7f}$$

Obviously, the sets (B7a, B7b, B7c) for amplitude quadratures and (B7d, B7e, B7f) for phase quadratures are independent and can be separated.

Using (3.3) and (3.14a) one can derive equations (3.15) for sum and difference amplitude quadratures $\beta_{a\pm}$.

Appendix C: Standard Quantum Limit

Here we discuss formula (3.25) and present details on derivation of SQL for force acting on mechanical oscillator.

We start from single-sided power spectral density (3.22) of signal force quadrature $f_{s\,a}$ of normalized signal force (3.10, 3.11), acting during time τ. Approximate condition of detection is

$$f_{s\,a} = \frac{F_{s0}}{\sqrt{2}\sqrt{2\hbar m\omega_m}} \geq \sqrt{\int_0^{2\pi/\tau} S_f(\Omega)\,\frac{d\Omega}{2\pi}} = \tag{C1}$$